

THE SCREW DRILL: IoT ZOMBIE FIRE MONITORING SYSTEM

WHAT ARE ZOMBIE FIRES?

Fire Migrating Underground



Spring



Resurfacing



Winter

REQUIRES



IoT Alerts to empower nearby residents & Indigenous response. [O-2.1, O-2.2]

Early detection of deep peat smoldering to prevent CO2 emissions and protect soil layers for forest regeneration. [O-1.3]



Autonomous, 12-month monitoring in isolated regions to identify hidden subsurface threats for government action. [O-1.1, O-3.1, O-3.2]

DESIGN CONCEPT

Signaling to communities and authorities

Prototyped ✓



InfluxDB (Database)



High Speed
Scalable



Grafana (Live Visualization)



Multi-Display



Python/
MicroPython
(Sensor/Comms Interface)

Reasons



Design Complexity



Existing Designs

Sensors



BME680

Monitor



Temperature



Air Quality



SKU
SEN0114

Monitor

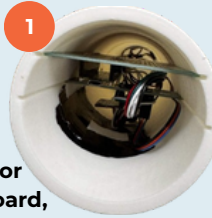


Soil Moisture

PROTOTYPE SYSTEM

Structural Subsystem

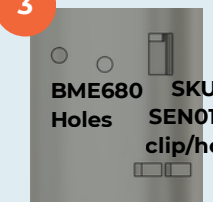
Electrical Subsystem



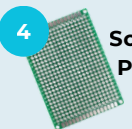
Slots for Perfboard, Inner Circuit, mesh fabric



Sleeve + Cross-pin Screw Joints



BME680 SKU
Holes SEN0114 clip/hole



Soldering + Perfboard



2-layered sensors

Depth Gradient of Data

When? Mass-deployed after initial firefighting work

Why?

IoT system monitors high risk areas

How? Screwed into the earth

DESIGN PERFORMANCE AND VERIFICATION

[S-1]

Verified BME680, SKU SEN0114 covers Temperature, Air quality, Soil Moisture



[S-2]

Validates accuracy monitoring temperature $\pm 1^\circ\text{C}$



[S-3]

Instantly picks up moisture level change $[+1000 \text{ ADC}]$



[S-4,5]

Data reading for 8cm, 49cm depth, 41cm span with 2-layered sensors collected



[S-6]

Data successfully transmitted via MQTT + WiFi



[S-7]

Reliable connection maintained up to 1m apart



[S-8]

Full dis(re)assemble with screw completed in <8 minutes

